

RITAM ROY

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PROFESSIONAL SUMMARY

Computer Science graduate and Junior AI Engineer with hands-on experience in Machine Learning, Data Science, and Business Analysis through industry roles, internships, and academic projects. Proficient in Python, Scikit-learn, Pandas, predictive modeling, data preprocessing, feature engineering, and exploratory data analysis. Passionate about Artificial Intelligence, Generative AI, Large Language Models (LLMs), Small Language Models (SLMs), and building robust, data-driven software solutions.

EXPERIENCE

ARBOOD CONSULTANCY PRIVATE LIMITED

June 2026 – Present

Junior AI Engineer

- Architecting and fine-tuning **Small Language Models (SLMs)** and **Retrieval-Augmented Generation (RAG)** pipelines for intelligent workflow automation and contextual question answering.
- Designing end-to-end AI workflows, integrating RESTful APIs for real-time model inference, and benchmarking evaluation metrics across custom domain datasets.

INTERNSHIP

T & D ENTERPRISE AND ENGINEERS

Dec 2025 – May 2026

Data Science Intern

- Engineered and deployed **Machine Learning & AI classification models** to accurately identify, score, and target high-probability convertible leads, significantly accelerating sales conversion rates.
- Designed and maintained automated **MS Excel and spreadsheet reporting dashboards** to track project milestones, lead pipelines, and operational KPIs for cross-functional teams.
- Conducted in-depth **Exploratory Data Analysis (EDA)** and data preprocessing on multi-source business datasets, transforming raw transactional data into actionable strategic insights.

PROJECTS

Rainfall Prediction Using Machine Learning

Python, Scikit-learn, Pandas, NumPy

- Developed an end-to-end precipitation forecasting model leveraging historical meteorological time-series data.
- Conducted extensive data preprocessing, missing value imputation, feature engineering, and exploratory data analysis.
- Evaluated multiple classification algorithms (Random Forest, Logistic Regression), achieving approximately **74% prediction accuracy**.

Chatbot Building Using SLM & RAG

Hugging Face, RunPod GPU, PyTorch, REST APIs

- Implemented an advanced **Retrieval-Augmented Generation (RAG)** pipeline to augment generative models with external factual knowledge, drastically reducing hallucination rates.
- Fine-tuned and deployed open-source Small Language Models from **Hugging Face** on **RunPod GPU** environments, optimizing throughput and memory utilization.
- Engineered REST API endpoints to facilitate real-time knowledge ingestion and seamless querying across curated Kaggle benchmark datasets.

TECHNICAL SKILLS

- Programming:** Python
- Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, PyTorch, Hugging Face Transformers
- Machine Learning & AI:** Linear Regression, Logistic Regression, Random Forest, Predictive Modeling, SLMs, LLMs, RAG, AI Workflows
- Data Science:** Data Cleaning, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Data Visualization
- Business Analysis & Tools:** MS Excel (Formulas, Pivot Tables, Dashboards), KPI Reporting, Requirement Gathering, Process Improvement
- Developer Tools:** Git, GitHub, Jupyter Notebook, Google Colab, RunPod GPU

EDUCATION

University of Engineering and Management, Kolkata

2020 – 2024

Bachelor of Technology in Computer Science and Engineering | CGPA: 7.84

Bidhannagar Government High School

2020

Higher Secondary (Science, WBCHSE) | Percentage: 70.20%

Bidhannagar Government High School

2018

Secondary (Class X, WBBSE) | Percentage: 78.00%